m = Cumulative frequency of the class preceding the median class

c = the class interval of the median class.

From the formula, it is clear that one has to find the median class first. Median class is, that class which correspond to the cumulative frequency just greater than $\frac{N}{2}$.

Example 5.17

The following data attained from a garden records of certain period Calculate the median weight of the apple

Weight in grams	410 – 420	420 – 430	430 – 440	440 – 450	450 – 460	460 – 470	470 – 480
Number of apples	14	20	42	54	45	18	7

Solution:

Weight in grams	**Number of apples**	**Cumulative Frequency**
410 – 420	14	14
420 – 430	20	34
430 – 440	42	76
440 – 450	54	130
450 – 460	45	175
460 – 470	18	193
470 – 480	7	200
Total	**N = 200**	

$$\frac{N}{2} = \frac{200}{2} = 100.$$

Median class is 440 – 450

$$\text{Median} = l + \frac{\frac{N}{2} - m}{f} \times c$$

$l = 440, \quad \frac{N}{2} = 100, \quad m = 76, \quad f = 54, \quad c = 10$

$$\text{Median} = 440 + \frac{100 - 76}{54} \times 10$$

$$= 440 + \frac{24}{54} \times 10 = 440 + 4.44 = 444.44$$

The median weight of the apple is 444.44 grams

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